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Research article

Comparison of simulation-based interprofessional education and video-enhanced interprofessional education in improving the learning outcomes of medical and nursing students: A quasi-experimental study

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ABSTRACT

Background: Use of interprofessional education has greatly expanded and is widely used to foster interprofessional collaborative practice competency in health professionals, especially during medical emergencies. Identifying the interprofessional education delivery mode that can maximize learning outcomes within available resources is crucial for achieving the sustainability of an interprofessional education program.
Objectives: To examine the learning outcomes of simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education modules and to analyze the variable costs between these two interprofessional education modules.
Design: A prospective quasi-experimental study.
Setting: A medical university in Northern Taiwan.
Participants: 24 medical students and 48 nursing students enrolled in a 4-week simulation-based interprofessional education or video-enhanced interactive discussion interprofessional education program.
Methods: Students' medical task performance, critical medical task performance, team behavior performance, and interprofessional collaboration attitude were examined at the pretest and posttest by using an objective structured checklist and team performance rating scale with a hands-on simulation model and structured questionnaires. The variable costs for simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education were estimated upon the completion of the course.
Results: All learning outcomes significantly improved for both the groups from the pretest to posttest. After the intervention, the simulation-based interprofessional education group exhibited significantly higher scores in medical task performance, critical medical task performance and team behavior performance than the video-enhanced interactive discussion interprofessional education group at the posttest. No significant difference was noted in interprofessional collaboration attitude between the groups. Cost analysis revealed that the simulation-based interprofessional education program was approximately two times more expensive in terms of staffing costs and required 1.3 times greater space charges than the video-enhanced interactive discussion interprofessional education program.
Conclusion: Both simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education modules effectively improved medical and nursing students' medical task performance, critical medical task performance and team behavior performance. The simulation-based interprofessional

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education group had more favorable learning outcomes, but the associated costs were higher. Educators should consider both learning outcomes and cost when choosing a design strategy simulation-based interprofessional education vs. video-enhanced interactive discussion interprofessional education for interprofessional education module development.

1. Introduction

Effective teamwork and collaboration among health-care professionals is key for high-quality and efficient healthcare management and is associated with improved patient care and outcomes (Rosen et al., 2018; Schmutz et al., 2019). Interprofessional collaborative practice is critical for patients with emergency medical conditions (Hazwani et al., 2020; Sarcevic et al., 2012). Healthcare providers must develop resuscitation competency and teamwork skills to address the ever-changing health-care environment, especially in the early phases of medical and nursing education in academic settings. Interprofessional education is being increasingly integrated into university-based teaching programs as an interactive teamwork educational model that provides opportunities for team members with different professional backgrounds to learn about, with, and from each other (Aldriwesh et al., 2022; Reeves et al., 2016). Cumulative evidence has indicated that interprofessional education can help students develop resuscitation competency and acquire skills required for successful teamwork (Onan et al., 2017). However, challenges related to interprofessional education delivery methods and available resources significantly affect the initiation and implementation of interprofessional education (Sunguya et al., 2014). The interprofessional education delivery module that can maximize learning outcomes within available resources remains to be identified. Moreover, limited related information is available for low/middle income countries with most reports from high income countries.

2. Background

Emerging studies have supported the positive effects of active learning on medical and nurse education and have indicated that active learning can enhance the retention of knowledge and skill over time (Berkhout et al., 2018; Bristol et al., 2019). In particular, optimal knowledge retention is required in active learning models, such as simulation education and group discussion, for learners to acquire and apply knowledge and skills on building teamwork behavior and collaboration (Berkhout et al., 2018; Bristol et al., 2019). In a systematic review of 23 studies, including 5231 students, authors reported that most interprofessional education courses were provided using small group discussions (15.9 %) or simulation sessions (13.6 %) to strengthen interprofessional education within an undergraduate curriculum (Berger-Estilita et al., 2020). Recently, multimedia technologies, such as video-enhanced teaching, are being increasingly used in medical and nurse education because it provides convenient educational resources for learners as well as sufficient and repetitive learning opportunities (Dong and Goh, 2015; Rose, 2020; Yousef et al., 2014). A hybrid curriculum blending the video-enhanced module with interactive discussion was determined to be a beneficial e-learning strategy for interprofessional education for university students (Prasad et al., 2020). Therefore, investigating the effects of interprofessional education based on simulation and video-enhanced interactive discussion on building teamwork behavior and collaboration skills can inform the development of the interprofessional education curriculum for undergraduate students.

Various challenges regarding interprofessional education are encountered in different countries, including those related to the design of the curriculum and the availability of adequate resources (Sunguya et al., 2014). The simulation-based approach is expensive and requires adequate resources for ensuring the sustainability of the program (Maloney and Haines, 2016; Zendejas et al., 2013). A cost analysis of the simulation-based approach and other available interprofessional education strategies is crucial for preparing, developing, and conducting feasible interprofessional education programs. Existing studies conducting cost analysis have mostly analyzed fixed costs, such as the purchase of mannequins and equipment, instead of examining variable costs, the cost of conducting the course, including staffing costs, space charges, and consumables, which are required for developing the course framework (Maloney and Haines, 2016; Zendejas et al., 2013). Furthermore, few studies have compared costs and learning outcomes among different interprofessional education learning models.

The present study (1) examined the learning outcomes of simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education to determine the approach that would result in more effective learning outcomes in medical task performance, critical medical task performance, team behavior performance, and interprofessional collaboration attitude and (2) analyzed the associated variable cost of each module.

2.1. Theoretical framework

The interventions in this study were developed on the basis of Kolb's experiential learning theory and Bandura's social learning theory (Bandura and Walters, 1977; Kolb et al., 2014). Kolb's experiential learning theory indicates that learner's knowledge is developed through the process of experience transformation, which consists of four key stages: concrete experience, reflective observation, abstract conceptualization, and active experimentation (Kolb et al., 2014). According to social learning theory, learners develop new behaviors by acquiring direct experience, by observing others' behavior within a social context, or by observing the consequences of others' behavior. In 1977, Bandura and Walters suggested four crucial processes that contribute to successful learning: attention, retention, reproduction, and motivation. Adopting these theoretical underpinnings, the research team designed and implemented two interprofessional education courses, namely simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education modules, for medical and nursing students. In each interprofessional education module, medical and nursing students interacted with each other in a team setting to manage hands-on simulated scenarios or discuss video scenarios to learn from and with each other. Students were oriented for the purpose of the course, which facilitated connecting the course content to patient care for maintaining their motivation and attention. Under the team learning model, students observed each other's behavior and understood the related consequences. The course progressed from easy to more difficult clinical situations in accordance with an educational scaffolding plan for both groups.

3. Methods

3.1. Study design

A prospective quasi-experimental design with pre- and post-test assessment (Fig. 1) was used at a medical university in Northern Taiwan from February to May 2020. The present study was approved by the institutional review board of the conducting institution.

3.2. Participants and setting

A total of 24 medical and 48 nursing students were enrolled into the

2020 interprofessional education course. The medical students were in their fifth year of a 6-year medical school program and they were participating in a clerkship rotation at either the emergency or internal medicine department of a tertiary teaching hospital. The nursing students were in the last semester of their 4-year Bachelor of Science in Nursing program and they were participating in clinical rotations at the same tertiary teaching hospital. All of the medical students were certified as advanced cardiac life support providers and all of the nursing students were certified as basic life support providers by the Taiwan Society of Emergency and Critical Care Medicine. Before the interprofessional education course, the research team conducted an instructional session to inform the medical and nursing students about the purpose and the process of the study. If they were not willing to participate in the study, they could still participate in the intervention session in order to receive the same instructional benefit. All of the students voluntarily agreed to participate in the study and all of them completed the study.

3.3. Interprofessional education course

The interprofessional education course was developed on the basis of the guideline recommended by interprofessional education collaborative (IPEC, 2016; Core Competencies for Interprofessional Collaborative Practice: 2016 Update (ipecollaborative.org), TeamSTEPPS 2.0 (Team Strategies and Tools to Enhance Performance and Patient Safety) established by the United States Agency for Healthcare Research and Quality (AHRQ, 2019; https://www.ahrq.gov/teamstepps/instructor/index.html), and the American Heart Association (AHA) guidelines for cardiopulmonary resuscitation. The core competencies were identical and consistent for both the 4-week simulation-based interprofessional education course and the 4-week video-enhanced interactive discussion interprofessional education course. The course consisted of three main parts: (1) critical patient management (assessment and management of the patient's airway, breathing, and circulation and the resuscitation of patients experiencing cardiac arrest); (2) teamwork skills (leadership, communication, mutual support, and situation monitoring); and (3) interprofessional collaborative practice (values and ethics for interprofessional practice, roles/responsibilities, and interprofessional communication).

The interprofessional education courses were developed and validated by an expert clinical panel, which consisted of physicians, nurse practitioners, registered nurses, and a physician internationally certified as a simulation educator (Certified Healthcare Simulation Educator). All of the clinical experts had at least 10 years of experience working in emergency departments and at least 5 years of simulation teaching experience. The clinical simulation scenario scripts were developed on the basis of the core content for both the simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education courses for hands-on practice. The research team adopted the same simulation scenarios for the development of interprofessional education video teaching materials.

During the course, the students were randomly assigned into the simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education groups (Fig. 1). For both the simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education groups, the students were then divided into four subgroups for teaching purposes. Each subgroup consisted of three medical students and six nursing students. In the simulation-based interprofessional education group, the students participated in a structured debriefing session after completing each hands-on simulation scenario. In the video-enhanced interactive discussion interprofessional education group, the students participated in a structured interactive discussion session and shared findings and thoughts after each video scenario. To maintain teaching consistency, all faculty members received training to provide step-by-step scripted guidance for both the simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education groups.

3.4. Outcome measurement

The study outcomes were medical task performance, critical medical task performance, team behavior performance and interprofessional collaboration attitude. All of the outcomes were examined at the pretest and posttest (i.e., 1 week after the completion of the 4-week course). To

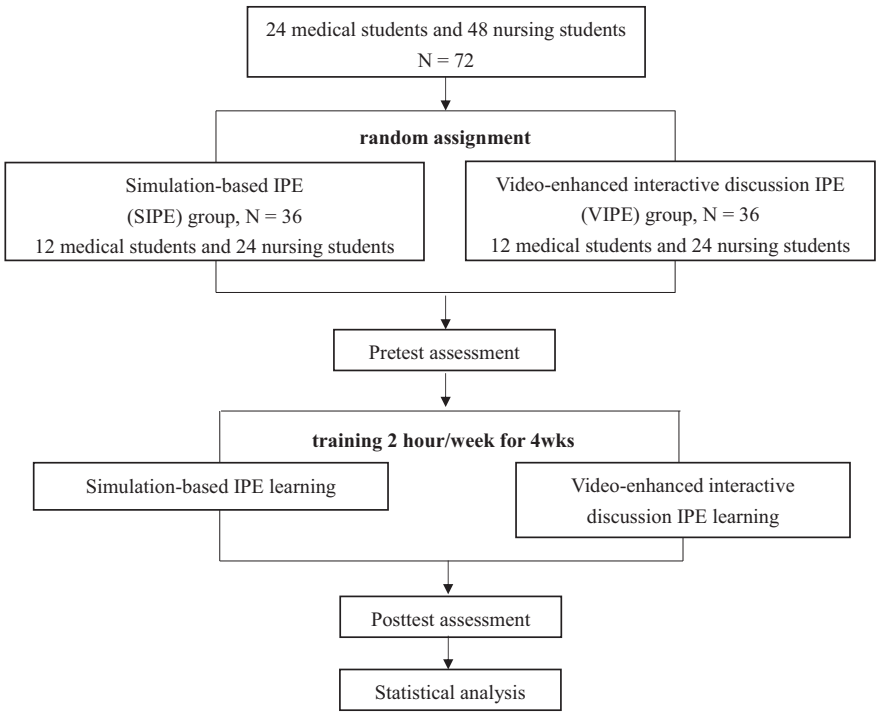

Fig. 1. Study flow diagram of Simulation-based IPE (SIPE) and Video-enhanced interactive discussion IPE (VIPE).

determine medical task performance, critical medical task performance, team behavior performance, each student team which consisted of one medical and two nursing students was rated by trained faculty during a standardized simulation scenario. The team consisted of the same teammates at the pretest and posttest. Data on cost analysis were collected and analyzed after course completion.

3.4.1. Management task performance

The Taiwanese 54-item rating checklist was used to examine a student team's task performance. This checklist was developed on the basis of the Taiwanese version of the American Heart Association guidelines for cardiopulmonary resuscitation, including initial assessment, care environment setting evaluation, airway management, patient reevaluation, and cardiac arrest management. Each item was rated using a dichotomous scale with 1 (Yes) and 0 (No), and the total score ranged from 0 to 56. The interrater reliability of this scale was 0.86 and 0.96 (mean 0.91) in the pretest and posttest, respectively, indicating a high level of agreement based on the criteria of ≥0.90 established by Graham et al. (2012).

3.4.2. Critical medical task performance

The 5-item Critical Medical Task Performance Scale was used to examine each student team's management of medical tasks in emergency situations. The research team developed this scale on the basis of resuscitation guidelines, and this scale was approved by the expert panel. The scale included the following parameters: 1) providing adequate oxygen in accordance with a patient's dynamic condition; 2) identifying cardiac arrest in a timely manner and providing high-quality cardiopulmonary resuscitation; 3) identifying shockable rhythms with the delivery of an in-time shock of the correct energy level; 4) administering resuscitation medicines correctly; and 5) activating local- and facility-level rapid response teams adequately. This checklist was rated on a dichotomous scale with 1 (complete) and 0 (incomplete), with a total score ranging from 0 to 5.

3.4.3. Team behavior performance

The 8-item Mayo High Performance Teamwork Scale was used to assess the student team's team behavior performance (Malec et al., 2007). Each item was assigned a score from 0 (never or rarely) to 2 (consistently), with the total score ranging from 0 to 16. The interrater reliability of this scale was 0.72 and 0.82 (mean 0.77) in the pretest and posttest, respectively, which is an acceptable level of interrater reliability (Graham et al., 2012).

3.4.4. Interprofessional collaboration attitude

The 13-item Interprofessional Collaboration Scale was used to assess interprofessional collaboration attitude. The original scale consisted of 26 items pertaining to three main aspects: communication, accommodation, and isolation (Kenaszchuk et al., 2010). The research team adopted and translated 13 items to develop the Taiwanese version of the scale on the basis of medical and nursing professional backgrounds. The students' responses for each item were rated on a 4-point Likert scale, with 1 indicating strongly disagree and 4 indicating strongly agree with the total possible score ranging from 13 to 52, with a higher score indicating a greater level of interprofessional collaboration attitude. The Cronbach alpha for this scale was 0.82.

3.5. Data collection procedure

Four raters who were certified as advanced cardiac life support providers by the Taiwan Society of Emergency and Critical Care Medicine were trained using a standardized process to ensure the accuracy of the assessment. For pretest rater calibration, each rater scored a standardized simulation scenario video independently. Raters then reviewed ratings together with a rating partner to discuss variances in scoring against expert ratings. These review sessions were facilitated by a member of the research team to provide clarity on how to rate specific elements and to reduce variability in the scoring of observed behaviors. Once the raters achieved an understanding of the correct rating for standardized scenarios, they reviewed and rated pretest and posttest events and scored them independently. The rating teams reviewed initial scores for each scenario and reached a consensus on the final combined rating for each event. The consensus score was used to compare outcomes within and between the groups as well as to provide the most accurate possible feedback to participants.

3.6. Data analysis

Categorical variables are presented as frequency and percentage. Continuous variables are presented as mean and standard deviation. The chi-squared test was performed to examine between-group differences in most demographic characteristics, except age, at the pretest. Due to the inclusion of small sample sizes with non-normal distribution, the Mann–Whitney test and Wilcoxon signed-rank test were performed to examine between-group and within-group differences, respectively. All data analyses were performed using IBM SPSS version 22 (IBM, Armonk, NY, USA). Statistical significance was determined using a two-tailed test with a P value of <0.05.

4. Results

A total of 72 students completed all of the tests and courses. Demographic characteristics were similar between the simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education groups (Table 1). At the pretest, the mean scores of all of the outcome measures were similar between the two groups (Table 2). At the posttest, the simulation-based interprofessional

Table 1
Demographic characteristic between groups ($N = 72$).

Variable	Simulation-based IPE (SIPE)	Video-enhanced interactive discussion IPE (VIPE)	P
	n (%)	n (%)	
Professional background			1.00
Nursing student	24 (66.7)	24 (66.7)	
Medical student	12 (33.3)	12 (33.3)	
Age (years), Mean (range)	22.50 (21–25)	22.33 (21–28)	0.28[a]
Gender			0.30
Male	12 (33.3)	8 (22.2)	
Female	24 (66.7)	28 (77.8)	
Experience of BLS			0.30
No	8 (22.2)	12 (33.3)	
Yes	28 (77.8)	24 (66.7)	
Experience of ACLS			1.00
No	23 (63.9)	23 (63.9)	
Yes	13 (36.1)	13 (36.1)	
Experience of team training			1.00
No	25 (69.4)	25 (69.4)	
Yes	11 (30.6)	11 (30.6)	
Experience of IPE training			0.91
No	25 (69.4)	31 (86.1)	
Yes	11 (30.6)	5 (13.9)	
Experience of resuscitation			0.78
No	28 (77.8)	27 (75.0)	
Yes	8 (22.2)	9 (25.0)	
Experience of ICU or ER			0.81
No	23 (63.9)	24 (66.7)	
Yes	13 (36.1)	12 (33.3)	

Note: IPE, interprofessional education; Chi-Square Test.
[a] Mann–Whitney U test.

Table 2
Descriptive statistics and independent-sample *t*-test of posttest and change between groups.

Outcome	Pretest		Posttest		Change in score, pretest to posttest		*P*
	M ± SD	*P*	M ± SD	*P*	MD ± SD	*P*[a]	
MTP		0.49		0.003*			0.01*
SIPE	14.42 ± 5.27		39.33 ± 5.21		24.92 ± 6.56	0.002*	
VIPE	13.08 ± 3.42		30.67 ± 6.58		17.58 ± 5.55	0.002*	
CMTP		0.32		0.01*			0.03*
SIPE	0.08 ± 0.29		2.00 ± 0.85		1.92 ± 0.10	0.003*	
VIPE	0.00 ± 0.00		1.00 ± 0.95		1.00 ± 0.95	0.010*	
TBP		0.81		0.03*			0.12
SIPE	4.75 ± 2.18		10.58 ± 2.43		5.83 ± 3.30	0.002*	
VIPE	5.08 ± 3.04		8.17 ± 2.86		3.08 ± 3.75	0.025*	
ICA							
Total score		0.72		0.24			0.26
SIPE	38.86 ± 4.98		43.03 ± 5.62		4.17 ± 5.05	<0.001*	
VIPE	38.11 ± 4.73		41.34 ± 3.78		3.14 ± 4.45	<0.001*	
Communication		0.44		0.07			0.26
SIPE	14.25 ± 2.08		16.06 ± 2.24		1.81 ± 2.49	<0.001*	
VIPE	13.92 ± 2.08		15.09 ± 1.54		1.11 ± 2.07	0.001*	
Accommodation		0.77		0.33			0.92
SIPE	15.47 ± 2.46		16.97 ± 2.44		1.50 ± 2.01	<0.001*	
VIPE	14.94 ± 1.91		16.40 ± 2.09		1.46 ± 1.87	<0.001*	
Isolation		0.76		0.44			0.44
SIPE	9.14 ± 1.78		10.00 ± 1.93		0.86 ± 2.46	0.006*	
VIPE	9.25 ± 1.48		9.86 ± 1.31		0.57 ± 1.65	0.052	

Note: MTP, Medical task performance; CMTP, Critical medical task performance; TBP, Team behavior performance; ICA, Interprofessional collaboration attitude; M, Mean; SD, Standard Deviation; SIPE, Simulation-based interprofessional education; VIPE, Video-enhanced interactive discussion interprofessional education.
* $P < 0.05$, two-tailed; Mann–Whitney U test.
[a] P: within group.

education group had significantly higher scores than did the video-enhanced interactive discussion interprofessional education group for medical task performance (39.33 ± 5.21 vs. 30.67 ± 6.58; $P = 0.003$), critical medical task performance (2.00 ± 0.85 vs. 1.00 ± 0.95; $P = 0.01$), and team behavior performance (10.58 ± 2.43 vs. 8.17 ± 2.86; $P = 0.03$). No significant differences in the interprofessional collaboration attitude score were noted between the simulation-based interprofessional education group and video-enhanced interactive discussion interprofessional education group in the posttest (43.03 ± 5.63 vs. 41.34 ± 3.78). Moreover, no significant differences were observed in all of the subdomains of interprofessional collaboration attitude, namely communication, accommodation, and isolation, between the groups. The improvement from the pretest was significantly higher in the simulation-based interprofessional education group than in the video-enhanced interactive discussion interprofessional education group for medical task performance (24.92 ± 6.56 vs. 17.58 ± 5.55; $P = 0.01$) and critical medical task performance (1.92 ± 0.10 vs. 1.00 ± 0.95; $P = 0.03$). No significant differences were noted in the improvement of team behavior performance (5.83 ± 3.30 vs. 3.08 ± 3.75) and interprofessional collaboration attitude (4.17 ± 5.05 vs. 3.14 ± 4.45) between the simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education groups.

Table 3
Comparison of the variable cost of simulation-based IPE and video-enhanced interactive discussion IPE learning.

Description	Simulation-based IPE (SIPE)	Video-enhanced interactive discussion IPE (VIPE)
Total Staffing cost[a]	172,800 NTD[b]	90,400 NTD[b]
Total Space charges	96,000 NTD[b]	72,000 NTD[b]
Total Consumables per course	10,000 NTD[b]	1000 NTD[b]
Total variable cost-	278,800 NTD[b]	163,400 NTD[b]

[a] Total faculty working hours (SIPE 160 h, VIPE 88 h) × an hourly wage for faculty (1000 NTD/h) + total faculty assistant working hours (SIPE 64 h, VIPE 12 h) × and hourly wage for faculty assistant (200 NTD/h); IPE, Interprofessional education.
[b] New Taiwan dollar (NTD).

A within-group comparison indicated that the students in both the groups exhibited improvement in multiple areas after the intervention. In the simulation-based interprofessional education group, significant improvements were observed for medical task performance (14.42 ± 5.27 to 39.33 ± 5.21; $P = 0.002$), critical medical task performance (0.08 ± 0.29 to 2.00 ± 0.85; $P = 0.003$), team behavior performance (4.75 ± 2.18 to 10.58 ± 2.43; $P = 0.002$), and interprofessional collaboration attitude (38.86 ± 4.98 to 43.03 ± 5.62; $P < 0.001$). In the video-enhanced interactive discussion interprofessional education group, significant improvements were observed for medical task performance (13.08 ± 3.42 to 30.67 ± 6.58; $P = 0.002$), critical medical task performance (0.00 ± 0.00 to 1.00 ± 0.95; $P = 0.010$), team behavior performance (5.08 ± 3.04 to 8.17 ± 2.86; $P = 0.025$), and interprofessional collaboration attitude (38.11 ± 4.73 to 41.34 ± 3.78; $P < 0.001$). In addition, a subgroup analysis was conducted between the nursing and medical students for interprofessional collaboration attitude. The interprofessional collaboration attitude of the nursing students significantly improved in the simulation-based interprofessional education group (43.08 ± 5.25 to 47.42 ± 5.49; $P = 0.034$), but not in the video-enhanced interactive discussion interprofessional education group (42.00 ± 4.86 to 42.29 ± 3.61). Improvement of the nursing students' interprofessional collaboration attitude was significantly higher in the simulation-based interprofessional education group than in the video-enhanced interactive discussion interprofessional education group (4.33 ± 7.23 vs. 0.92 ± 4.72, $P = 0.04$) (Supplementary Table 1).

For cost analysis, the research team analyzed the variable costs including the costs of staffing, space, and consumable goods in the simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education groups (Table 3). The total faculty working hours were 160 h in the simulation-based interprofessional education group and 88 h in the video-enhanced interactive discussion interprofessional education group. The total working hours of the assistants were higher in simulation-based interprofessional education group than in the video-enhanced interactive discussion interprofessional education group (64 h vs. 12 h). The total staffing cost was approximately two times higher in the simulation-based interprofessional education group than in the video-enhanced

interactive discussion interprofessional education group. The total space cost was higher in the simulation-based interprofessional education group than in the video-enhanced interactive discussion interprofessional education group (New Taiwan Dollar, NTD $ 96,000 vs. NTD $ 72,000). The details of the cost analysis are reported in Supplementary Table 2.

5. Discussion

The results of this study indicated that both simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education significantly improved medical task performance, critical medical task performance, team behavior performance, and interprofessional collaboration attitude in medical and nursing students. Compared with the video-enhanced interactive discussion interprofessional education module, the simulation-based interprofessional education module appeared to be more effective overall. However, simulation-based interprofessional education had higher costs in terms of faculty working hours, faculty assistant working hours and educational space requirements than did video-enhanced interactive discussion interprofessional education. These findings provide important information that can inform design choices for an interprofessional education program and provide a framework for future interprofessional education implementation for medical emergency learning for undergraduate students.

In the present study, the research team observed that the simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education programs focusing on medical emergency circumstances were effective in improving the students' medical task performance and team behavior performance. By contrast, previous studies have reported that different interprofessional education teaching methods that did not focus on team performance and interaction under medical emergency scenarios had no additional positive effect on the development of teamwork knowledge, attitudes, or behavior (Clay-Williams et al., 2013; Fox et al., 2018). Interprofessional collaborative practice involving emergency medical care situations is more complex than interprofessional collaborative practice involving nonemergency medical situations (Gharaveis et al., 2018); this difference might affect students' team coordination and effectiveness. Findings of the present study supported that the utilization of simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education as interprofessional education teaching methods can facilitate the development of the ability of medical and nursing students to manage emergency medical conditions.

In the current interprofessional education study, the course duration was 2 h per week for 4 weeks. Previous studies including a 1-day course have reported no significant learning outcome (Clay-Williams et al., 2013; Reeves et al., 2016), indicating an effect of this course timing and duration on various outcomes between studies. Although results of the present study indicated that the implementation of an interprofessional education of 2 h per session extended over 4 weeks is more effective than a single 1-day interprofessional education workshop, future studies are warranted because of limited evidence related to how interprofessional education course timing and duration impacts immediate and sustained learning outcomes (Marlow et al., 2017; Reeves et al., 2016).

Data from the present study indicated that the simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education programs exerted positive effects on the undergraduate medical and nursing students. Several studies have reported that the learning level of specific participant cohorts vary; thus, results must be interpreted in the narrow context of each learning group (e.g., undergraduate students, postgraduate students, or clinical practitioners) because the identical teaching strategy may result in different learning outcomes in learners of different learning levels (Aldriwesh et al., 2022). Additional studies should investigate the differential effects of an identical teaching strategy on students with different learner levels.

Many studies have suggested that different learning models of interprofessional education can improve students' attitude, which is a Kirkpatrick level one outcome (Marlow et al., 2017; Reeves et al., 2016). Results from the present study indicated that both the simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education modules exerted a positive effect on the medical students' interprofessional collaboration attitude; however, only the simulation-based interprofessional education program demonstrated positive outcomes on the nursing students' interprofessional collaboration attitude. Findings of the present study suggest that the video-enhanced interactive discussion interprofessional education learning model is effective for increasing interprofessional collaboration attitude among medical students at a lower cost. To develop nursing students' interprofessional collaboration attitude, simulation-based learning is more effective although it is more costly. Future studies are warranted to explore the factors underlying these findings.

Simulation-based interprofessional education is an effective learning method because it provides hands-on opportunities for students to develop and practice new skills. The limitation of this learning method is that it is conducted through a face-to-face model with higher instructor, equipment, and space costs. Results of the present study revealed that a well-designed video-enhanced interactive discussion interprofessional education program can improve student's medical task performance, critical medical task performance and team behavior performance. This type of learning model can be implemented through cloud-based learning without in-person contact. This finding indicates that video-enhanced interactive discussion interprofessional education can serve as a possible alternative to simulation-based interprofessional education, results in effective learning and has the potential to reduce in-person contact which might be perceived as an advantage in the post-pandemic world (Rose, 2020).

In the present study, the research team observed that for the same number of students, simulation-based interprofessional education had approximately two times higher staffing costs and 1.33 times higher space charges than video-enhanced interactive discussion interprofessional education. Simulation-based learning models have demonstrated better outcomes than traditional methods, but require increased faculty working hours and space charges (Cook et al., 2012; Maloney and Haines, 2016; Zendejas et al., 2013). Use of interprofessional education for development of team behavior and the ability to perform medical tasks is critical among healthcare students managing medical emergency scenarios. This importance is emphasized by the fact that the senior undergraduate medical and nursing students will soon be a part of the primary care team upon graduation and need these essential skills to provide high quality patient care. Simulation-based interprofessional education provided greater improvement than video-enhanced interactive discussion interprofessional education for medical task performance, critical medical task performance, and team behavior performance for the medical and nursing students in this study. This finding suggests that the additional cost of simulation-based interprofessional education can provide higher-level behavioral change outcomes, which are the level 3 learning outcomes of the Kirkpatrick program evaluation model (Kirkpatrick, 2009). This is crucial when improved teamwork and collaboration behaviors are targeted outcomes for managing emergency scenarios because these skills are related to patient safety and clinical care outcomes.

6. Limitations

This study has several limitations relevant to the interpretation of results. The number of participants and type of healthcare professionals was limited. Students who participated in this study had variable experiences in medical clerkship and nursing rotations, which may impact their performance. Our participants were prelicensure students; thus, caution should be exercised when attempting to apply these findings to

postgraduate students and practicing professionals.

7. Conclusion

Both the simulation-based interprofessional education and video-enhanced interactive discussion interprofessional education modules are effective in improving the medical task performance, critical medical task performance, team behavior performance and interprofessional collaboration attitude of teams composed of medical and nursing students. The learning outcomes for simulation-based interprofessional education are superior to those for video-enhanced interactive discussion interprofessional education in educating students to manage emergency scenarios, although it is accompanied by higher costs.

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CRediT authorship contribution statement

Jen-Chieh Wu: Conceptualization, Methodology, Software, Supervision, Writing – original draft, Writing – review & editing, Resources, Funding acquisition. **Hui-Wen Chen:** Visualization, Investigation, Writing – review & editing. **Yu-Jui Chiu:** Visualization, Investigation, Data curation. **Yi-Chun Chen:** Writing – review & editing. **Yi-No Kang:** Writing – review & editing, Software, Formal analysis. **Yi-Ting Hsu:** Visualization, Investigation, Data curation. **John M. O'Donnell:** Conceptualization, Methodology, Software, Supervision, Writing – original draft, Writing – review & editing, Resources, Funding acquisition. **Shu-Yu Kuo:** Conceptualization, Methodology, Software, Supervision, Writing – original draft, Writing – review & editing, Resources, Funding acquisition.

Declaration of competing interest

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.

This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue.

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